

Track 52: SolidWorks Mechanical Design

June 2026

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Tier 3 — Specialized | Departments: Mechanical · Mechatronics

Skill path: Sketch → Part Modeling → Assembly → Simulation → Sheet Metal → Drawings → PDM

Related: [Track 37 — CAD/CAM](#) · [Track 45 — CATIA](#) · [Track 46 — ANSYS](#)

Track Overview

Who this is for: Mechanical and Mechatronics students targeting design engineer roles in manufacturing, automotive, and product companies.

Prerequisites: Engineering drawing and mechanics basics.

Tools: SolidWorks (latest), Simulation module, Composer (intro), PDM vault (6-month+).

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	SolidWorks literacy	Part + assembly + drawing
1 Month	80–100	Junior designer	Mechanism design project
3 Months	250–300	Design engineer intern	Simulated + documented product
6 Months	500–600	CSWP-ready portfolio	5+ industry-style projects

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Interface, sketch tools, constraints	5 part models
D3–4	Assembly mates, exploded views	Mechanism assembly
D5–6	Drawings, GD&T intro, BOM	Manufacturing drawing set
D7–8	Simulation: static stress, FOS	FEA report
D9–10	Capstone	Complete design package

1-Month Foundation

Week	Topics	Project
W1	Advanced modeling: sweeps, lofts, surfaces	Complex parts library
W2	Top-down design, configurations, design tables	Parametric product family
W3	Sheet metal, weldments	Fabrication-ready designs
W4	Motion study, rendering, capstone	Product design project

3-Month Job Ready

Machine elements (gears, shafts, bearings) + assembly design + simulation validation + design portfolio.

6-Month Professional

CSWP certification prep + surfacing + mold design intro + PLM/PDM workflow + client-style capstone.

Assessment Rubric

Weekly Quiz 20% | CAD labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Tier 3 brochure name **“SolidWorks”**. For CAM/manufacturing add [Track 48 CNC](#). For FEA depth add [Track 46 ANSYS](#).